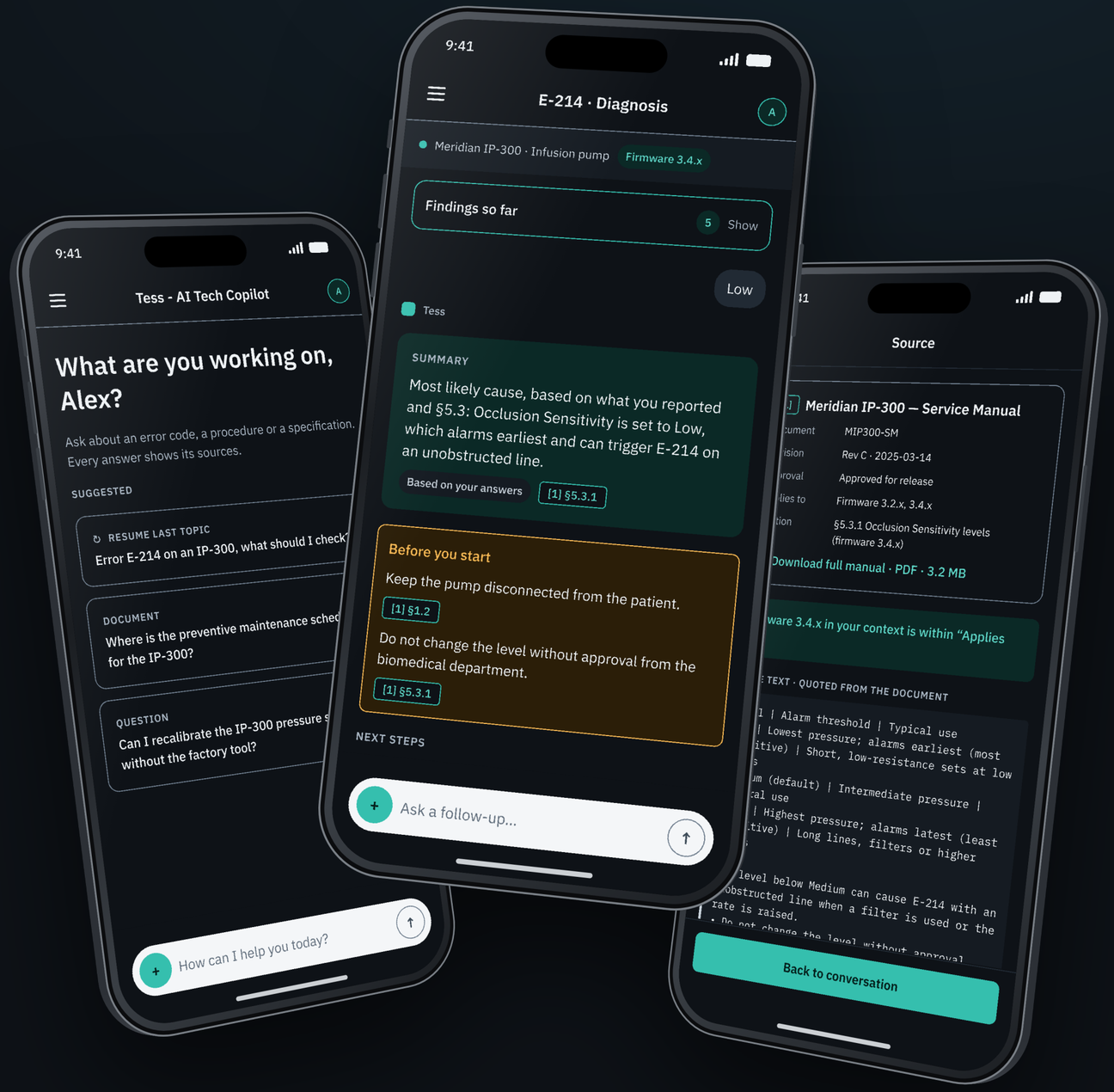
RADIX DESIGN CHALLENGE

Tess
AI Tech Copilot

An AI assistant that helps field engineers find, understand and verify technical information, built mobile first.

Yuri Saback · Senior Product Designer

Prototype | Figma | PRD



SITUATION

Alex needs the right answer in the field, and the proof behind it



Alex, 34

Field Service Engineer

Installs, inspects and repairs medical equipment in hospitals and clinics, moving between sites.

- Uses the phone to consult information on site
- Switches between equipment, manuals and specialist support
- Must know when the documents are not enough

Design hypothesis, not research

The problem

Manuals, specifications, error codes and troubleshooting guides live in different places. Finding the right one takes time, and the person needs it while standing at the equipment.

Job to be done

"When I need to inspect, maintain or troubleshoot a medical device, I want to find the relevant technical information and understand the next steps, so I can work efficiently while following authorized procedures."

Design decision: mobile first

Alex works in many places and the phone is the fastest tool at hand. I designed for the phone first. The prototype also has a desktop version, a static preview, but the focus is mobile.

Defined in the PRD · sections 1 and 2

TASK

Design a copilot people can trust, not just one that answers

FIND

One question, many documents

Searches manuals, specs and procedures from a single query.

INTERPRET

A structured answer

Summary, warnings, steps, for this equipment.

VERIFY

Evidence one tap away

Shows the source, says what it could not substantiate, points to support.

Five risks to design against

- Unsupported or fabricated answers
- Outdated documentation
- Wrong equipment context
- Overreliance on AI
- Insufficient evidence

Behind all five: the AI agent must be very well structured, so its research stays coherent with the project and with the user who is searching.

Boundary

The assistant supports the professional. It never authorizes a procedure or replaces manufacturer instructions.

Brief

A responsive, interactive prototype with at least 3 final screens, evaluated on reasoning, UX structure, simplification, visual quality, work with AI and autonomy.

Defined in the PRD · sections 3, 5, 6 and 9

ACTION 1 · REASONING

Simplify: one equipment, one flow, evidence at every step

01
Start
Suggested queries, no setup

→

02
Context
Equipment and problem

→

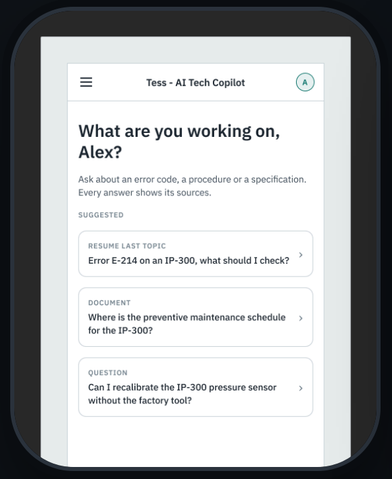
03
Answer
Structured, with uncertainty

→

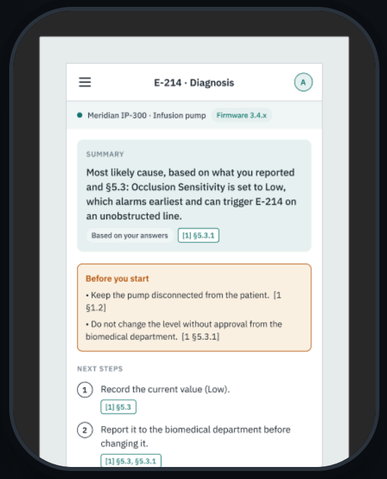
04
Verify
Open the cited source

→

05
Continue or escalate
Follow-up or authorized support



Wireframe · Main
Reviewed before visual design



Wireframe · Response
Hierarchy decided first

A guided diagnosis, not a data dump

Fictional Meridian IP-300 infusion pump, error E-214. Tess asks short questions until the likely cause appears, so the user stays in control.

One scripted ending

The demo cannot fail: each turn has one active action, so every evaluator reaches the same complete story.

Mobile first, desktop kept

Wireframes for mobile (M1–M7) and desktop (D1–D2). The mobile flow is fully testable; desktop is a static preview.

Figma · 01 Flow Figma · 02 Wireframes

ACTION 2 · DECISIONS

Eight decisions, each tied to a risk

DECISION	WHY
Tess asks for the firmware instead of assuming it	A wrong equipment context gives a wrong procedure
A citation on every step, inside the text	Verify without leaving the answer; no separate source list
Warnings come before the steps	A warning below the steps gets missed
Source text and AI explanation in separate blocks	Never blend a quote with a paraphrase
"Ruled out" is labelled "not verified by the assistant"	Show the limits of what was checked
Insufficient evidence: no steps, escalation first	Never fill a gap with plausible steps
Dark mode only, contrast checked	One theme fully verified beats two half-verified
No hover, 44 px primary targets	Touch first; pressed state instead of hover

All of these decisions follow the hypotheses, requirements, risks and guardrails defined in the PRD created at the start of the project (sections 4, 8, 9 and 10).

ACTION 3 · DESIGN SYSTEM

Built once, in Figma and in code, with the same names

4
variable collections: Primitives, Color (Light and Dark), Spacing, Radius

10
text styles in IBM Plex Sans and Mono

35
components with variants, reused in every final screen

25
text and boundary pairs checked against WCAG 2.2 AA



Direction

Graphite & Teal, picked from three MedTech proposals. Calm, technical, high contrast for hospital light.

Type

IBM Plex Sans

`Plex Mono for quoted source text`

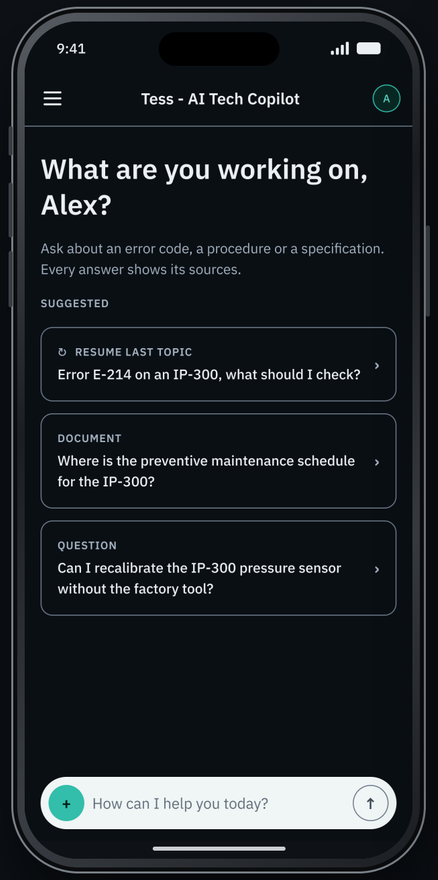
Parity with code

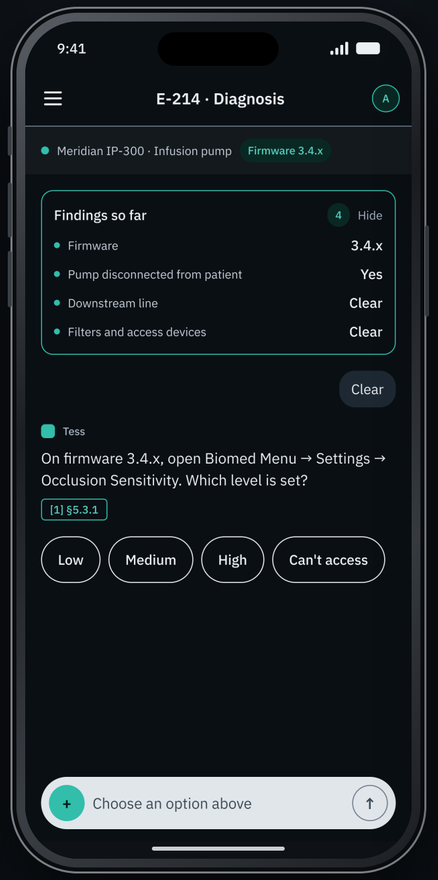
Figma variables carry code syntax and mirror `tokens.css`, so design and prototype cannot drift.

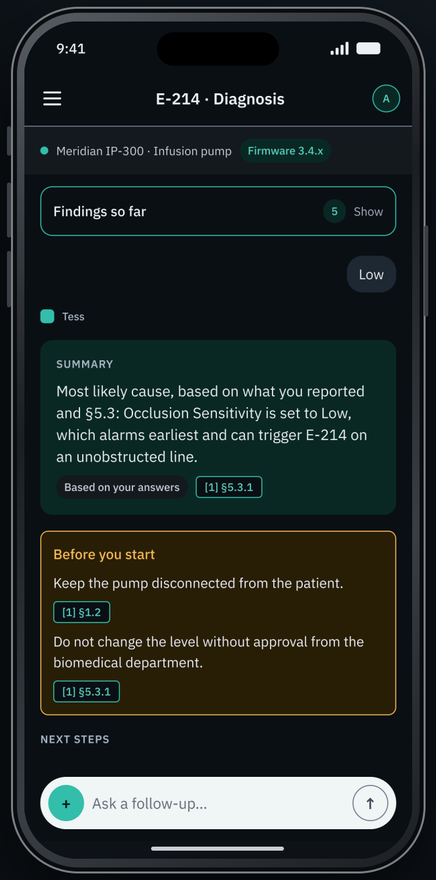
Figma · 04 Foundations Figma · 05 Components

ACTION 4 · FINAL SCREENS 1 TO 3

From start to a structured answer







1 Main interface

No equipment assumed. Suggested queries and a ready message field.

2 AI interaction

Tess asks for the firmware and what the pump shows, and records each answer.

3 Structured response

Summary, warnings first, a citation on each step, what was ruled out and what is not covered.

Desktop version also available

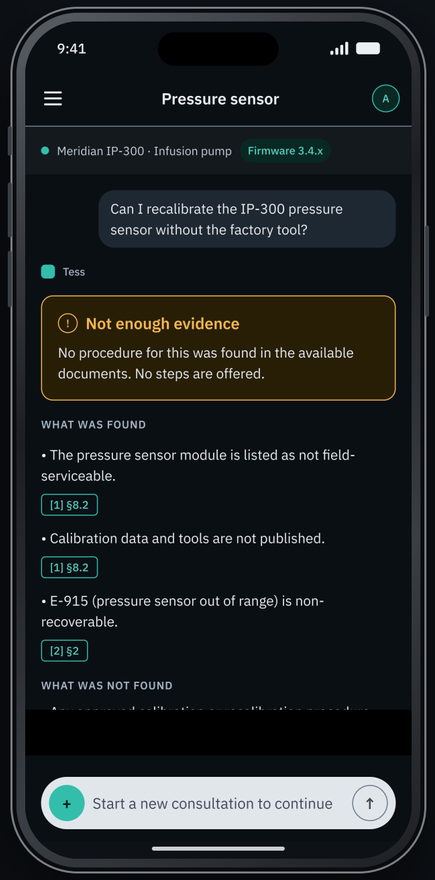
The prototype includes a desktop version, a static preview that is not interactive. Use the "View desktop version" button under the phone. This presentation focuses on mobile, the tested experience.

Figma · 06 Final screens Try the prototype

ACTION 5 · FINAL SCREENS 4 AND 5

Verify, or admit the limit





4 Source detail

- Document, section, revision, approval
- Firmware applicability check
- Quoted text apart from the AI explanation

The third strategic screen: traceability is the concept's differentiator and the direct answer to the fabrication and overreliance risks.

5 Insufficient evidence

- What was found, with citations, and what was not found
- No steps, no invented procedure
- Primary action: contact authorized support

A variant of the response screen, built from the same components.

Figma · 06 Final screens Try the prototype

ACTION 6 · AI AND AUTONOMY

Directed by me, built with AI

Definition
PRD, persona, scope

Wireframes
Approved flow

Design system
Variables, components

Prototype
Mobile first

Deploy
GitHub and Vercel

High fidelity
5 final screens

Handoff
States and specs

Every phase ended with my review and approval before the next one started.

What I decided

- The scenario and the one scripted ending
- The visual direction and the mobile-first approach
- Behavior rules: citations, warnings first, no hover
- Every correction after reviewing the prototype

What AI executed

- Figma file through its MCP connection
- Prototype code, versioned and deployed
- Accessibility and contrast checks
- Review of the work against the PRD

11
states mapped in the handoff flow

11
specification panels: behavior, motion, accessibility

1
live link, updated on every push to main

Figma · 07 Developer Handoff GitHub

RESULT

What was delivered, and what is not proven

Delivered

- 5 final screens in 2 scenarios
- Design system and developer handoff
- Testable mobile prototype, live
- Review against the PRD success criteria, requirements, risks and guardrails

Limits and conscious choices

- No interviews or usability tests: nothing is claimed about time saved or trust
- Desktop is a static preview, available in the prototype through the "View desktop version" button; mobile is the tested experience
- "Simulated content" is flagged outside the phone so the product looks real
- All manuals and equipment are fictional

Next

- Test H1 to H4 with field engineers
- Real, governed documentation
- Interactive desktop
- Measure answer verification and escalation

Prototype | Figma | PRD | Source code

Yuri Saback · Senior Product Designer